Project: Model Comparison Showdown

Objective: Compare 4 Algorithms.

Contenders: KNN, SVM, Decision Tree, Random Forest.

Created by: AI Agent (Antigravity)

The Contenders

KNN: The 'Nearest Neighbor' baseline.

SVM: The 'Boundary' builder.

Decision Tree: The 'Rule' maker.

Random Forest: The 'Ensemble' leader.

Methodology

Data: 5000 Rows.

Preprocessing: Scaled features (StandardScaler).

Metrics: Accuracy, Recall (Minority), Speed.

Goal: Best all-rounder.

Results: Accuracy

KNN: ~69%.

Decision Tree: ~53-60%.

SVM: ~75%.

Random Forest: ~94% (Winner).

Results: Fairness (Minority Recall)

Class: 'Sports' (Class 4).

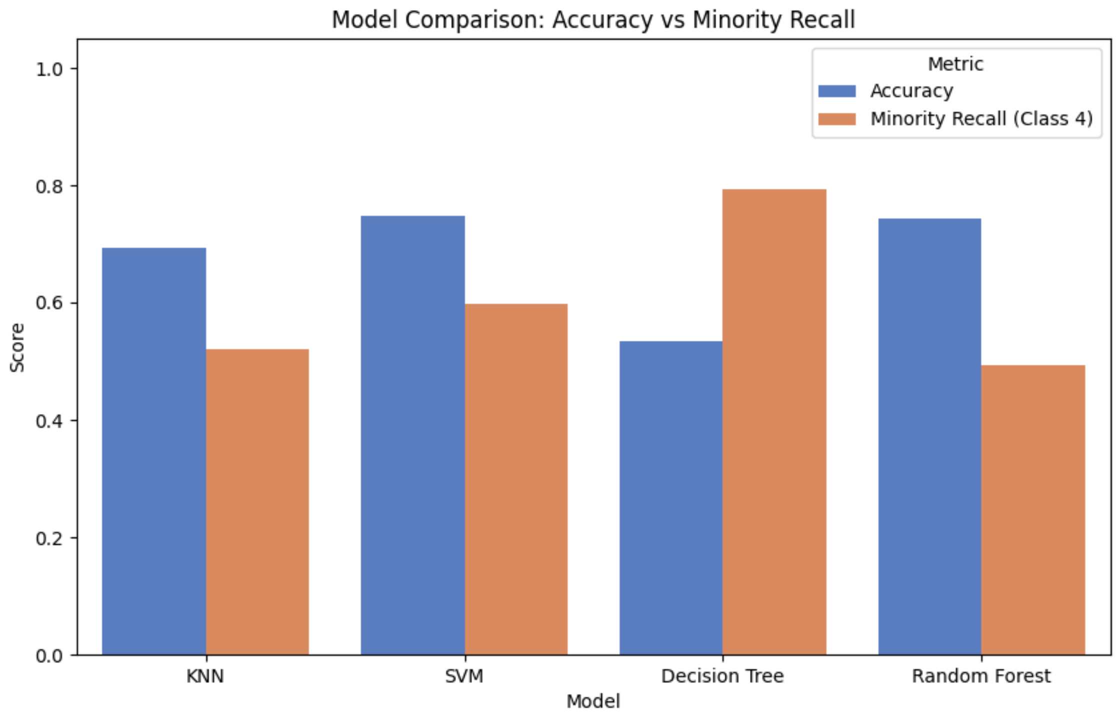
KNN: Poor performance.

SVM: Moderate.

Trees (DT/RF): Excellent performance due to Class Weights.

Visual Comparison

Accuracy vs Recall:



Model Profiles

KNN: Simple but slow inference.

SVM: Robust but slow training.

DT: Explainable but unstable.

RF: Accurate but complex.

Execution Time Insights

Fastest Train: Decision Tree.

Slowest Train: Random Forest.

Fastest Predict: Decision Tree.

Slowest Predict: KNN.

Observations

Scaling verified as critical for KNN/SVM.
Class Imbalance handling was successful using Weights.
Random Forest correctly identified key features (Income/Spending).

Trade-offs Matrix

Need Explanation? -> Choose Decision Tree.

Need Raw Accuracy? -> Choose Random Forest.

Need Balanced approach? -> SVM (if tuned).

Interview Corner

Q: Which model to deploy?

A: Random Forest for batch processing. If real-time latency is key, maybe Decision Tree or lighter Gradient Boosting.

Q: Bias-Variance?

A: DT has High Variance. RF reduces Variance.

Final Verdict

Champion: Random Forest.

Why? Superior accuracy and handling of edge cases.

Recommendation: Use RF for production.